

2-zero

$$\begin{cases} X_{3j} = X_{1j} - X_{14} & j = 5, \dots, n-1 \\ X_{3n} = -X_{14} \\ X_{2j} = X_{3j} + X_{24} & j = 5, \dots, n \end{cases} \Leftrightarrow \begin{cases} X_{2n} = X_{3n} + X_{24} \\ = -X_{14} + X_{24} \end{cases} \quad \begin{array}{l} \text{if } j < n: \\ X_{2j} = X_{3j} + X_{24} \\ = X_{1j} - X_{14} + X_{24} \end{array}$$

choose maximal set of independent variables s.t. we can send the rest to ∞ and get rid of constraints

- ① if X_{24} is indep. then we must have X_{3j}, X_{2j} go to ∞ $j = 5, \dots, n$
 then $X_{14} \rightarrow \infty$
 $\rightarrow$ we keep X_{1j} $j = 5, \dots, n-1$ as indep.

$$(X_{24}, X_{15}, X_{16}, \dots, X_{1n-1})$$

- ② if X_{2n} is indep. then X_{1j}, X_{2j} must be ∞ ($j = 4, \dots, n-1$)
 $X_{3n} \rightarrow \infty$, we keep X_{3j} $j = 5, \dots, n-1$

$$(X_{2n}, X_{35}, X_{36}, \dots, X_{3n-1})$$

- ③ send both X_{24} and $X_{2n} \rightarrow \infty$, also $X_{14}, X_{3n} \rightarrow \infty$

for each $j = 5, \dots, n-1$ we must make a choice:

- if X_{1j} is indep: X_{3j} not be ∞ , also $X_{2j} \rightarrow \infty$

note: the eq: $X_{2j} = X_{1j} + X_{2n}$ cannot be obtained after taking the ∞ limits because the eqs we need now are $\infty = \infty$.

- if X_{3j} is indep: X_{1j} not be ∞ , also $X_{2j} \rightarrow \infty$

- if X_{2j} is indep: $X_{3j}, X_{1j} \rightarrow \infty$

$\rightarrow$ for each $j \in \{5, \dots, n-1\}$ choose one of (X_{1j}, X_{2j}, X_{3j}) as indep.

All the terms that have chords chosen from both fix U a set of independent are killed.

Conversely, anything that has chords from outside these cases must survive.

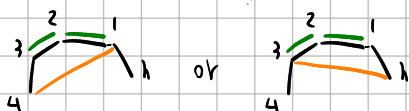
A) $(X_{24}, X_{15}, X_{16}, \dots, X_{1n-1})$

B) $(X_{2n}, X_{35}, X_{36}, \dots, X_{3n-1})$

C) one of each (X_{1j}, X_{2j}, X_{3j}) , $j = 5, \dots, n-1$

What survives:

- anything that has 14 or $3n$



- if it has 24 needs to survive case (A)

- $2n$

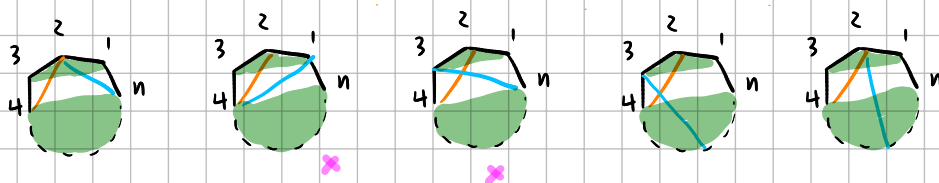
- 14

- $3n$

- $3j$ $j = 5, \dots, n-1$

- $2j$ $j = 5, \dots, n-1$

→ things that are not in $(X_{24}, X_{15}, X_{16}, \dots, X_{1n-1})$



- if it has $2n$ needs to survive case (B)

will have one or more of

- 24

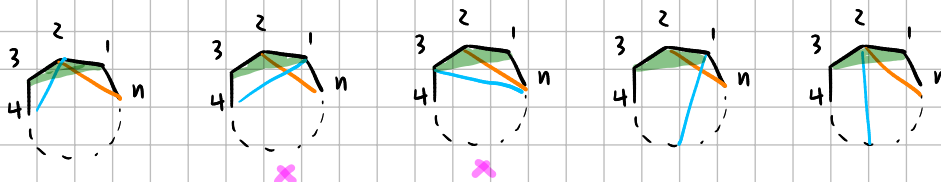
- 14

- $3n$

- $1j$

- $2j$

$j = 5, \dots, n-1$



- if there is $2j$ need to survive case (C)

- 24

- $2n$

- 14

- $3n$

- $1j$

- $3j$

- both $(1e, 2e)$ $e \neq j$

- both $(1e, 3e)$ $e \neq j$

- both $(3e, 2e)$ $e \neq j$

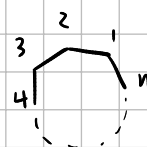
$$\boxed{n=6} \begin{cases} X_{35} = X_{15} - X_{14} \\ X_{36} = -X_{14} \\ X_{25} = X_{35} + X_{24} \\ X_{26} = X_{36} + X_{24} \end{cases}$$

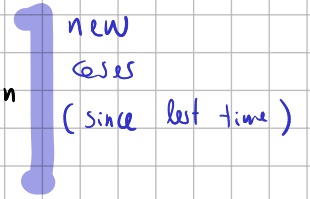
$$X_{35} = x - \lambda_1$$

$$X_{36} = -\lambda_1$$

$$X_{25} = x - \lambda_1 + \lambda_2$$

$$X_{26} = -\lambda_1 + \lambda_2$$





- 14
- 3n
- both $1j$ and $3j$ for some $j \in \{5, \dots, n-1\}$

→ if nothing shows out of $(n+1)$ -gen then it will be contained in a $(n+2)$ gen